

4-8-8-4 “Big Boy” Assembly

NOTE: Sanding to fit pieces together will be required. The amount of sanding will depend on how many shells you use in MakerWare - I used 2. Parts were designed without tolerances. Sanding to fit WILL BE REQUIRED. Parts were also made to fit on the MakerBot Replicator 2.

1. Track

14 “Rail Bed Sections” are required

Assemble 6 sections with short side of the bed ties to the outside. (This will give you the proper gauge when setting the wheels.)

2. Front Engine Frame

- I. Assemble Front Engine Frame according to picture ([Need Isometric and side view JPGs](#))
- II. Assemble Main Drive Wheels, Mount “68 inch Tire” on wheels with front of the tire flush to the rim of the wheel (not with the edge of the counterweight)
- III. Insert corresponding “Crank Pin” in “Main”, “Front/Back” and “Intermediate” wheels. Set the correct depth using the corresponding rods.
- IV. Place 2 “Spring Set” pieces back to back on each of the 8 main axles. (DO NOT glue at this time)
- V. Glue wheel assemblies on to the axles, using a section of the track to set the proper distance between the wheels. Be sure to have the wheels in the same position (counter weights down).
- VI. Assemble “Link and Trunion”, Radius Bar, Link Support Frame. Set aside. (easier to mount assembled)
- VII. Add Wheel/Axle Assemblies to the frame.(see pictures for proper order).
- VIII. Glue “Spring Set” flush to outside edge of frame.
- IX. Add “Front Rod”, “Intermediate Rod” and “Back Rod” to wheels for proper orientation of wheels
- X. Place assembly on track bed to align all the wheels and center the frame on the axles.
- XI. When happy with orientation of wheels. Glue axles in place.
- XII. Add linkage(Main Rod, “Link and Trunion”, “Crosshead & Guide” “Piston Rod”, etc.) from the inside out. Dry fit pieces... Wheel orientation will determine final position of much of the linkage.

3. Front Engine Truck

- I. Assemble “Engine Truck Swing Frame”, “Engine Truck Spring Replacement,” and 2 “Engine Truck Side Frames.” Set aside.

- II. Place 2 Engine Truck Axle Boxes on each of the Engine Truck Axles. Glue "36 inch Wheel" to each axle using the track assembly to set width.
 - III. Attach axles to the frame assembly, setting axle blocks flush to edge of frame (see picture)
 - IV. Glue Engine Truck Assembly to Front Frame Assembly using track section to align wheel assemblies. Glue in place.
 - V. Set aside front assembly.
4. Rear Engine
 - I. Repeat all steps for "Front Engine" assembly (Section 2)
5. Trailing Truck
 - I. Place "42 inch Wheels" on Axles
 - II. Sand to fit Axle Bearings on Axles
 - III. Glue Trailing Truck Frame in halves....left and right sides.
 - IV. Fit Axles into frame halves and glue frame together.
 - V. Use track to align wheels and center frame...glue wheels in place
 - VI. Use track again to align Trailing Truck Frame with Rear Engine Assembly. Glue.
 - VII. Set aside Rear Engine Assembly.
6. Smoke Box/Boiler/ Cab
 - I. Assemble Smoke Box and Boiler Cylinder sections individually.
 - II. Glue sections together aligning either the top or side seams.
 - III. Assemble Cab Section...floor to walls. Then glue roof sections together.
 - IV. Align Floor/Wall Assembly to Roof Assembly and glue together.
 - V. Attach Sandbox #3 to rear section of Boiler.
 - VI. Attach Cab Assembly to rear of Boiler. (Sandbox and angle of floor supports should level the cab. Sand to fit if necessary)
 - VII. Attach Cab Support pieces to floor of Cab Assembly.
 - VIII. Attach other Sandboxes, Dome and Smoke Stack
 - IX. Apply "Seam Plate" over all Seams (Cut to fit...and builder's discretion)
 - X. Set aside Boiler Assembly
7. Pilot
 - I. Center Pilot on Front Engine Assembly.
 - II. Glue Radiator Core assembly to top of frame behind Pilot (see picture)
 - III. Glue Air Pumps to either side of Radiator Core. Align hoses.
 - IV. Attach Air Pump Shield to top of Pilot.
 - V. Place Sump Radiator behind Air Pump Shield.
8. Front and Rear Engine Assembly
 - I. Place Front and Rear Engine assemblies on the tracks.
 - II. Slide the Rear Engine forward over the end of the Front Engine.
 - III. Align the pivot holes. (If Rear Engine wheels lift of the track, sand frame to allow the wheel to return to level.)

- IV. Attach "Pipe T" to Front Engine.
- V. Fit "Steam Pipe – Back Engine" between Pipe T and the front of the back cylinder. (If Steam Pipe does not reach/fit to the back engine, Move Rear Frame accordingly. Glue frames together!!!)
- VI. Attach both Steam Pipe – Back Engine to assembly.

9. Boiler Attachment

- I. Place Boiler Assembly on top of Frame Assemblies.
- II. To get the proper location of the Boiler Assembly on the Frame assembly...Use the pipe "Steam Pipe Front Engine". This pipe will run from the mounting flange on the bottom of the Smoke Box to the mating flange on the frame just behind the area where the Radiator Core is mounted.
- III. Check for interference...Seam Plates may need to be trimmed to allow proper seating on the frame.
- IV. When satisfied with the location, glue Boiler to Frame.
- V. Glue the "Steam Pipe Front Engine" to the 2 flanges used when locating the Boiler

10. Walkways and additional Steam Pipes.

- I. Assemble 4 Walkways and glue to Boiler.
- II. Assembly "Steam Pipe – Boiler", "Steam Pipe T Pipe" and "Steam Pipe – Bent." (Align seams to help keep parts level.)
- III. Use cutout in Front Walkway to position pipe.
- IV. Glue Pipe assembly to both the Smoke Box and the top of the Rear Cylinders while keeping the front end of the Boiler Pipe horizontal.
- V. Attach "Steam Pipe L/R" and "Steam Pipe Elbow" between the Steam Pipe Boiler and the Front Cylinder.

11. Front Step Assembly

- I. Use Front Engine Assembly
- II. Attach "Front Step Bracket – FRONT" to front of Front Step Plate
- III. Attach Front Step Bracket Seat L/R to matching Front Step Bracket L/R
- IV. Glue Front Step to frame behind Radiator Core.
- V. Glue Front Step Brackets L/R to both the Cylinder (part of Frame) and the back edge of the Front Step.
- VI. Attach Bumper Steps to both sides from Pilot to underside of Front Step Plate (See Pic)
- VII. Attach Pilot Steps 1 and 2 to corresponding sides of pilot.
- VIII. Attach Pilot Step Railing and Pilot Light Railing to Pilot Step.
- IX. Mount Pilot Light on Base over the Radiator Core.

12. Railings and Other Parts.

- I. Assemble horizontal parts of Hand Rail in 2 parts...the 4 parts of New Handrail and New Handrail 2.

- II. Place Hand Rail Supports on New HandRail, 2.5 inches from cab....then every 3 inches on center. Up to the Smoke Box. Align all supports and glue.
- III. On New Handrail 2...starting 1 inch back from edge, position and align supports every 3.5 inches.
- IV. Using locomotive...align vertical part of Handrail to front of Front Runboard and New Handrail 2 with supports approx. 90 degrees to Smoke Box surface.
- V. Align and attach the 2 Handrail sections, aligning supports.
- VI. Begin at Cab to attach Handrail Assembly. Cut to fit the supports at the Sandboxes.
- VII. At Smoke Box, Handrail will need to be heated slightly to conform to the step down.
- VIII. Attach Ladders to Cab Floor
- IX. Attach Bell and Bell Crib to top of Smoke Box
- X. Attach Indicator Lamps to either side of the Bell, align to roughly the top of the Bell Crib.